

ДЗ по биоинформатике

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1 Задание 1

Приведем сначала параметры запуска и результаты поиска:

BLAST® » blastp suite » results for RID-XHYPJUWG014

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Job Title **TP53 tumor antigen non-redundant search**

RID **XHYPJUWG014** Search expires on 04-11 21:27 pm Download All ▾

Program **BLASTP** Citation ▾

Database **nr** See details ▾

Query ID **NP_000537.3**

Description **cellular tumor antigen p53 isoform a [Homo sapiens]**

Molecule type **amino acid**

Query Length **393**

Other reports Distance tree of results Multiple alignment MSA viewer ?

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to E value to Query Coverage to

Filter Reset

Рис. 1: Параметры запуска

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
✓	cellular tumor antigen p53 isoform X1 [Gorilla gorilla gorilla]	Gorilla gorilla gorilla	814	814	100%	0.0	99.75%	408	XP_063556659.1
✓	Chain K, Cellular tumor antigen p53 [Homo sapiens]	Homo sapiens	814	814	100%	0.0	100.00%	395	TXZZ_K
✓	Chain C, Cellular tumor antigen p53 [Homo sapiens]	Homo sapiens	814	814	100%	0.0	100.00%	417	8R1F_C
✓	cellular tumor antigen p53 isoform a [Homo sapiens]	Homo sapiens	813	813	100%	0.0	100.00%	393	NP_000537.3
✓	mutant tumor protein p53 [Homo sapiens]	Homo sapiens	813	813	100%	0.0	99.75%	393	AYE20617.1
✓	mutant tumor protein p53 [Homo sapiens]	Homo sapiens	812	812	100%	0.0	99.75%	393	AYE20613.1
✓	mutant tumor protein p53 [Homo sapiens]	Homo sapiens	812	812	100%	0.0	99.75%	393	AYE20623.1
✓	p53 antigen [Homo sapiens]	Homo sapiens	812	812	100%	0.0	99.75%	393	AAA61212.1
✓	cellular tumor antigen p53 isoform X2 [Gorilla gorilla gorilla]	Gorilla gorilla gorilla	812	812	100%	0.0	99.75%	393	XP_004058559.3
✓	p53 transformation suppressor, partial [Homo sapiens]	Homo sapiens	812	812	100%	0.0	99.75%	393	CAA42633.1
✓	tumor protein p53 (Li-Fraumeni syndrome), isoform CRA_c [Homo sapiens]	Homo sapiens	812	812	100%	0.0	99.75%	408	EAU90142.1
✓	tumor suppressor p53 [Homo sapiens]	Homo sapiens	812	812	100%	0.0	99.75%	393	AFN61606.1
✓	p53 [synthetic construct]	synthetic construct	811	811	100%	0.0	100.00%	403	UNB14053.1
✓	mutant tumor protein p53 transcript variant 1 [Homo sapiens]	Homo sapiens	811	811	100%	0.0	99.75%	393	WOX59026.1
✓	mutant tumor protein p53 [Homo sapiens]	Homo sapiens	811	811	100%	0.0	99.75%	393	AYE20618.1
✓	mutant tumor protein p53 [Homo sapiens]	Homo sapiens	811	811	100%	0.0	99.75%	393	AYE20622.1
✓	truncated mutant tumor protein p53 [Homo sapiens]	Homo sapiens	811	811	100%	0.0	99.75%	393	UIW38580.1
✓	mutant tumor protein p53 [Homo sapiens]	Homo sapiens	811	811	100%	0.0	99.75%	393	AYE20612.1
✓	mutant tumor protein p53 [Homo sapiens]	Homo sapiens	811	811	100%	0.0	99.75%	393	AYE20610.1
✓	tumor suppressor gene p53 [Homo sapiens]	Homo sapiens	811	811	100%	0.0	99.75%	393	BAJ52715.1
✓	tumor protein p53, partial [synthetic construct]	synthetic construct	811	811	100%	0.0	99.75%	394	AA42852.1

Рис. 2: Лучшие хиты

Ответы на вопросы:

- Первый найденный клеточный белок: cellular tumor antigen p53 isoform X1 [Gorilla gorilla gorilla]
- E-value = 0.00, т.е. в базе данных из случайных последовательностей мы встретим 0.00 последовательностей со скором выше, чем у последовательности белка TP53 в базе nr.

- Скор - очень относительная вещь и зависит от параметров запуска (матрицы замен (для АК), штрафов (для нуклеотидов)), от длины последовательности, и не говорит ничего о статистической значимости результата.

2 Задание 2

Приведем сначала параметры запуска и результаты поиска:

BLAST® » blastx » results for RID-XJ070ESG014

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Job Title **NM_000518:Homo sapiens hemoglobin subunit...**

RID **XJ070ESG014** Search expires on 04-11 21:53 pm Download All ▾

Program **BLASTX** Citation ▾

Database **swissprot** See details ▾

Query ID **NM_000518.5**

Description **Homo sapiens hemoglobin subunit beta (HBB), mRNA.**

Molecule type **rna**

Query Length **628**

Other reports ?

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to E value to Query Coverage to

Filter Reset

Рис. 3: Параметры запуска

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain; Contain...	Homo sapiens	301	301	70%	8e-105	100.00%	147	P68871.2
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Gorilla g...	Gorilla gorilla gorilla	300	300	70%	3e-104	99.32%	147	P02024.2
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Hylobat...	Hylobates lar	294	294	70%	5e-102	98.63%	146	P02025.1
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Semnop...	Semnopithecus s...	291	291	70%	4e-101	97.26%	146	P02032.1
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Colobus...	Colobus polykomos	291	291	70%	5e-101	95.92%	147	P19885.2
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Ateles p...	Ateles paniscus	290	290	70%	2e-100	95.92%	147	Q6WN22.3
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Ateles g...	Ateles geoffroyi	289	289	70%	3e-100	95.92%	147	P68232.2
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Macaca ...	Macaca fuscata f...	289	289	70%	4e-100	95.24%	147	P68222.2
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Pithecia p...	Pithecia pithecia	289	289	70%	4e-100	95.92%	147	Q6WN27.3
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Lagothrix...	Lagothrix lagotricha	289	289	70%	4e-100	95.92%	147	Q6WN25.3
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Chloroc...	Chlorocebus aeth...	288	288	70%	6e-100	95.89%	146	P02028.1
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Callithrix...	Callithrix jacchus	288	288	70%	8e-100	96.60%	147	Q6WN28.3
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Sapajus ...	Sapajus apella	288	288	70%	1e-99	95.24%	147	P67821.3
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Callimico...	Callimico goeldii	287	287	70%	2e-99	95.24%	147	Q6WN21.3
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Papio a...	Papio anubis	286	286	70%	6e-99	94.56%	147	Q9TSP1.2
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Saimiri s...	Saimiri sciureus	286	286	70%	8e-99	94.56%	147	P02036.2
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Ptilocobus...	Ptilocobus badius	285	285	70%	9e-99	95.21%	146	P02033.1
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Alouatta b...	Alouatta belzebul	285	285	70%	1e-98	95.24%	147	Q6WN29.3
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Mico arg...	Mico argentatus	285	285	70%	1e-98	96.58%	146	P18985.1
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Mandrill...	Mandrillus sphinx	285	285	70%	1e-98	94.52%	146	P08259.1
☒	RecName: Full=Hemoglobin subunit delta; AltName: Full=Delta-globin; AltName: Full=Hemoglobin delta chain (Pan tr...	Pan troglodytes	285	285	70%	1e-98	93.88%	147	P61772.2
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Leontoc...	Leontocebus fusc...	285	285	70%	2e-98	95.21%	146	P02039.1
☒	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain (Macaca ...	Macaca mulatta	285	285	70%	2e-98	94.52%	146	P02026.1

Рис. 4: Лучшие хиты

Ответы на вопросы:

- Нужно запустить BlastX, в условии задачи нам дана **мРНК**, а найти мы хотим гомологичные **белки**. Конечно, для этой задачи еще в теории подходит tBlastX, но задача 'найти гомологи' обычно не подразумевает поиска против уже определенных белковых последовательностей, а не теоретически возможных белков, синтезирующихся с нуклеотидных последовательностей. (для канонического протеома уже все рамки считывания известны)
- Запустил
- +3
- BlastX автоматически транслирует в 3 (6) рамках и ищет хиты. Руками тоже можно так сделать, но это очевидно будет менее удобно.

3 Задание 2

Приведем сначала параметры запуска и результаты поиска:

Job Title
Myoglobin blastp against swissprot

RID
[XJ1MTFAP016](#) Search expires on 04-11 22:18 pm [Download All](#) ▼

Program
BLASTP [Citation](#) ▼

Database
swissprot [See details](#) ▼

Query ID
[NP_005359.1](#)

Description
myoglobin isoform 1 [Homo sapiens]

Molecule type
amino acid

Query Length
154

Other reports
[Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#) [?](#)

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

[+ Add organism](#)

Percent Identity

E value

Query Coverage

to

to

to

Filter

Reset

Рис. 5: Параметры запуска

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Homo sapiens]	Homo sapiens	312	312	100%	2e-110	100.00%	154	P02144.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Hylobates agilis]	Hylobates agilis	311	311	100%	9e-110	99.35%	154	P62734.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Pan troglodytes]	Pan troglodytes	310	310	100%	2e-109	99.35%	154	P02145.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Gorilla beringei]	Gorilla beringei	310	310	100%	2e-109	99.35%	154	P02147.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Pongo pygmaeus]	Pongo pygmaeus	308	308	100%	2e-108	98.70%	154	P02148.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Papio anubis]	Papio anubis	300	300	100%	2e-105	96.10%	154	P68084.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Macaca fascicularis]	Macaca fascicularis	298	298	100%	9e-105	95.45%	154	P02150.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Rousettus aegyptiacus]	Rousettus aegyptiacus	295	295	100%	2e-103	94.16%	154	P02163.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Sus scrofa]	Sus scrofa	294	294	100%	6e-103	93.51%	154	P02189.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Tupaia glis]	Tupaia glis	288	288	100%	9e-101	91.56%	154	P02165.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Castor fiber]	Castor fiber	287	287	100%	2e-100	90.26%	154	P14396.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [feri]Orycteropus afer	feri]Orycteropus afer	287	287	100%	3e-100	90.26%	154	P02164.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Lutra lutra]	Lutra lutra	286	286	100%	6e-100	90.91%	154	P11343.3
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Callithrix jacchus]	Callithrix jacchus	286	286	100%	7e-100	90.91%	154	P02152.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Aotus trivirgatus]	Aotus trivirgatus	285	285	100%	1e-99	90.26%	154	P02151.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Sapajus apella]	Sapajus apella	285	285	100%	1e-99	90.91%	154	P02153.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Ochotona princeps]	Ochotona princeps	284	284	100%	4e-99	90.26%	154	P02171.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Didelphis virginiana]	Didelphis virginiana	284	284	100%	4e-99	90.26%	154	P02193.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Ctenodactylus guineensis]	Ctenodactylus guineensis	283	283	100%	1e-98	90.26%	154	P20856.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Lagostomus maxwelli]	Lagostomus maxwelli	283	283	100%	2e-98	88.96%	154	P04250.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Perodicticus potto]	Perodicticus potto	283	283	100%	2e-98	89.61%	154	P02166.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Proechimys guairae]	Proechimys guairae	282	282	100%	2e-98	88.96%	154	P04249.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Lagothrix lagotis]	Lagothrix lagotis	282	282	100%	3e-98	89.61%	154	P02154.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Erinaceus europaeus]	Erinaceus europaeus	282	282	100%	3e-98	89.61%	154	P02156.2
✓	RecName: Full=Myoglobin; AltName: Full=Nitrite reductase MB; AltName: Full=Pseudoperoxidase MB [Saimiri sciureus]	Saimiri sciureus	281	281	100%	6e-98	88.96%	154	P02155.2

Рис. 6: Лучшие хиты

Ответы на вопросы:

- 1e-32, 115, 43.15
- запустил
- 2e-27, 119, 41.61
- РАМ30 штрафует за несоответствие сильнее, а значит направлена на поиск более близких гомологов, чем BLOSUM62. Мы получили большую статистическую значимость для базы BLOSUM62, что говорит о том, что имеющийся у нас белок является дальним гомологом для danio regio, поэтому для его поиска больше подходит РАМ30. Это согласуется с биологическим контекстом задачи.

4 Задание 4

Приведем сначала параметры запуска и результаты поиска:

ⓘ Your search is limited to records include: **Drosophila melangaster (taxid:7227)**

Job Title	Searching orthologs of drosophila
RID	XJ3D98XH016 Search expires on 04-11 22:48 pm Download All
Program	BLASTP Citation
Database	swissprot See details
Query ID	NP_009225.1
Description	breast cancer type 1 susceptibility protein isoform 1 [Hom ...
Molecule type	amino acid
Query Length	1863
Other reports	Distance tree of results Multiple alignment MSA viewer

Filter Results

Organism only top 20 will appear ☐ exclude
 Type common name, binomial, taxid or group name
[+ Add organism](#)

Percent Identity to
E value to
Query Coverage to

Рис. 7: Параметры запуска

Sequences producing significant alignments									
Download Select columns Show 250									
☒ select all 1 sequences selected GenPept Graphics Distance tree of results Multiple alignment MSA Viewer									
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accessio
☒	RecName: Full=E3 ubiquitin-protein ligase RING1; AltName: Full=RING-type E3 ubiquitin transferase RING1; AltName: E...	Drosophila melan...	49.8	49.8	3%	3e-06	39.06%	435	Q9VB08.1

Рис. 8: Лучшие хиты

- ПОНЯЛ
- сделано
- E3 ubiquitin-protein ligase RING1;
- 3 процента, 39.6 процентов
- Консервативны только отдельные домены, отсюда можно сделать вывод, что белок BRCA имеет доменную структуру.